

Building Trust: Cryptocurrency & UX Strategy

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ABSTRACT

Despite the variety of coin options, from Bitcoin to Dogecoin, most coins do not survive the volatile market past their first year [7]. Understanding user experience and barriers to cryptocurrency adoption may contribute to growth potential. This paper discusses the significance of trust in user experience for cryptocurrency by exploring documented challenges for new and novice users compared to established users. Existing UX strategy and design recommendations for building trust are explored to show how developing a strategy that considers user trust is necessary to create cryptocurrency platforms and tools that can successfully secure new users.

KEYWORDS

Human-centered computing, Human-computer interaction (HCI) HCI theory, concepts, and models, blockchain, cryptocurrency

INTRODUCTION

Blockchain, also known as a distributed ledger, enables users to remain untraceable when completing electronic transactions [5]. It combines three technologies: (1) distributed databases, (2) encryption, and (3) consensus protocols [8]. This combination creates the opportunity to build applications around a representative shared state (e.g. a ledger)[9]. The term blockchain is rooted in how it functions as each transaction generates a record made up of data blocks, each encrypting the latter block[5]. Cryptocurrencies such as Bitcoin (a misnomer frequently used for all crypto coins) gave rise to this technology [5]. The benefits of cryptocurrencies extend beyond the anonymity of individual users or transactions.

According to Plansky, O'Donnell, & Richards (2016), cryptocurrency and blockchain have several benefits, including:

- Efficient bookkeeping and compliance through a centralized ledger that can track every transaction, create
- Improving business applications
- Supporting emerging economies that will not have to rely on existing financial institutions
- Growth through competition & new cryptocurrency-based services

Rapid growth can be seen in numerous services, platforms, and coins centered around cryptocurrency. Initially a peer-to-peer (P2P) concept [5,6], cryptocurrencies now have multiple forms of intermediaries such as exchange platforms, wallets, and payment processors, each charging a fee [6]. Companies exploring blockchain applications, include American Express, Barclays, IBM, and Nasdaq [5]. Applications of blockchain outside of finance are particularly significant in their potential to protect identities and prevent identity theft [5]. Relevant fields include the public sector (e.g., public records and election/voting), medical records, and commerce for large-scale retail purchases [5].

Despite its potential as a disruptor to standard financial institutions, less than a tenth of blockchain platforms survive their first year [7]. Additionally, engaging with cryptocurrency is not an easy task for current users and non-users [1]. The unfamiliarity of blockchain means a lack of understanding of its value and creates a poor user experience which then becomes a barrier to user entry [3]. Yet, in 2014 there were 483 cryptocurrencies available [5]. Even though new crypto coins continually pop up, such as Ethereum and Dogecoin, the technology faces low acceptance among consumers and business networks. The slow adoption and prevalence of new coins and crypto technology present a clear gap that has been unaddressed in strategy and design [2, 8]. Developing a strategy that considers user trust is necessary to establish cryptocurrency platforms and tools that can secure new users successfully.

CHALLENGES & RISKS

The gaps between hype over cryptocurrencies and blockchain possibilities and adoption rates are shaped by the challenges and risks of being a crypto user. Challenges are present externally and internally for crypto companies and their users. External challenges include a lack of clear regulation (beyond licensing) for cryptocurrencies and providers, price stability, and technical specifications (e.g., encryption, identity, and transaction standards)[5]. Market volatility also prevents investors, users, businesses, and governments from accepting and using cryptocurrency as a payment method [5]. Mass adoption is not feasible since cryptocurrency is not a mainstream payment method [5].

Also, there are risks associated with tools such as wallets where thefts can occur [6], or users can lose keys, and irreversible losses can be grave. The potential adverse outcomes are magnified because no central authority

monitors transactions [1]. Use context also affects risks, including how crypto assets are stored and their worth [1].

While these are constraints that individual companies cannot solve, they are related to broad issues when developing or using cryptocurrency. Understanding the varying external challenges is essential because they inform the internal challenges that crypto companies face.

According to Saez [2020], companies face internal challenges such as poor management practices and exploratory strategies for launching and developing blockchain-enabled platforms. Entrepreneurs may be motivated to create blockchain without proper consideration of returns, and larger businesses may set out to develop a blockchain-enabled platform without fully understanding the technology they are creating [7]. Additionally, much of blockchain is exploratory and without standardization across its varying applications, there is no standard strategy model to incentivize users and maximize value [7].

TRUST & USER EXPERIENCE

THE TRUST PARADOX

A central challenge for all parties is trust [6,7]. While Albayati [2020] describes trust as an external construct, it is still essential to garnering user trust for technology adoption. Trust is the main factor in consumers' behavior, decisions, and acceptance [2]. There are various forms and definitions of trust which can overlap at various levels[9]. One example is system-level through technology trust. Technology trust is based on understanding how a system works and predictability in system behavior [9]. More broadly, trust refers to how comfortable, confident, and secure consumers are when using a given technology. Specific to financial technology, trust is necessary for successful transactions in virtual and physical spaces [10]. Ideally, customers should have a low perception of risk probability, and trust should be high.

Interestingly, there is a paradox within trust assessments of blockchain. Trust in the blockchain is required from users, but the technology functions by excluding conventionally trustworthy intermediaries like governments and banks [8]. Yet, blockchain creates trust when there are concerns over data integrity, such as fears around record management [10]. Cryptocurrencies are high-risk investments [7] but inherently generate trust when users trust technology over institutions with the hope that blockchain can replace institutions [9]. Trust is said to be achieved in blockchain technology through its immutable and transparent ledger [11]. Essentially if the system can be trusted so can individual transactions because the system will provide security. Yet, how can users trust crypto coins or crypto platforms if they know regulation is lacking, coins can crash, or their digital wallets can be stolen?

Regulation is a frequent and appropriate answer [1, 2, 6] but it is a lengthy process unlikely to be established anytime soon. It is not feasible for companies to wait for regulation with growing competition and their desire to enter the market quickly. Although regulation is significant to user trust, design, perceived ease of use, experience levels, and social influence are important factors [2]. Abramova, Voskoboynikov & et al.'s [2021] survey of crypto-asset(e.g., cryptocurrency or digital tokens) users capture varying experience levels, motivations, and concerns. In order of expertise, cypherpunks are experienced advocates or enthusiasts of cryptocurrency, holders as traders and investors with primary concerns are profit and security, and rookies as inexperienced and motivated by fear of missing out [1]. Key differences were found across cryptocurrency ownership, storage, risk concerns, security practices, and demographics across each user group, with cypherpunks having at least six years of experience with cryptocurrency and fewer security concerns.

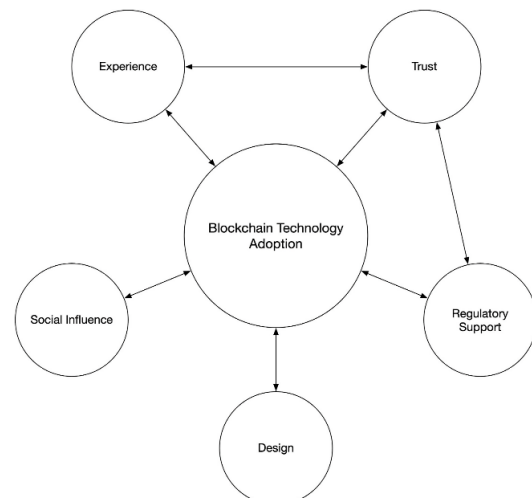


Figure 1. External constructs affecting blockchain technology adoption

In agreement with the study mentioned above, Albayati's [2020] findings show that trust, experience levels, and regulatory support reinforce one another (see Figure 1), and design is separated from this relationship, but good design and user-friendly interfaces had a significant impact on perceived ease of use. User-centered approaches contrast this model and connect design and trust directly by emphasizing design choices as essential to user experience. Additionally, design considerations for trust because user errors and security breaches can have irreversible consequences [1]. Potentially developing cryptocurrency systems that prevent user error could support or reinforce trust. Yet, user-centered design considerations are often left out of strategies that emphasize trust in blockchain technology including cryptocurrency platforms.

UX Frameworks & Design Solutions

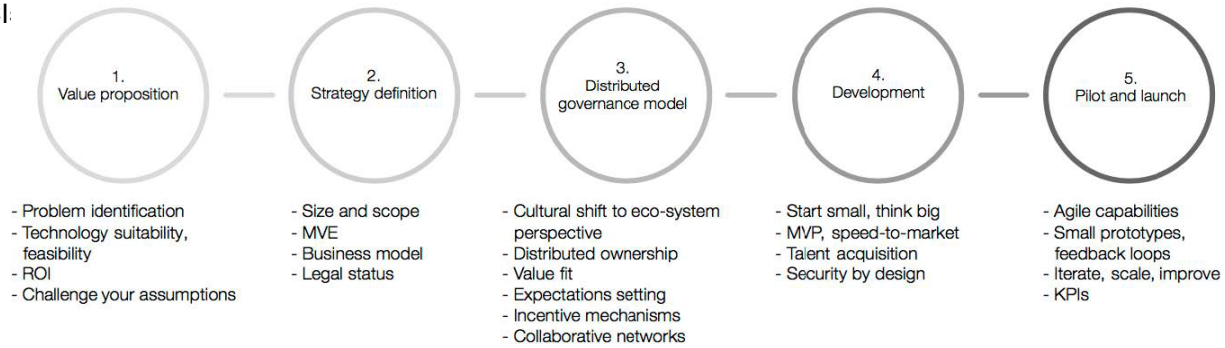


Figure 2. Saez's Success Factors

Trust is a part of user experience and broader goals. A circular relationship exists between trust and the design of a blockchain application—trust requirements influence the design of blockchain applications and the design influences trust [10]. Jang and Han [2020] developed a UX framework specifically for blockchain experiences based on general UX (usability, affect, and sociability) and technological UX (social value, economic value, and trust) to evaluate user experiences. General UX refers to essential aspects of the experience of any technology, and technological UX includes characteristics unique to blockchain experiences [3.] These categories are intended to be distinct without overlapping [3]. This framework addresses trust, but there is no description of how to align evaluation within product design phases or situate it within broader business goals.

Another approach for developing trust through UX focuses on design elements such as trust supporting design features (TSDEs) [10] and control mechanisms employed to reduce risks [9]. TSDEs are implemented to build trust at the application level and include the user interface, soft factors, labels, and information. Control measures minimize risk by drawing on the existing user community, such as reviews and ratings. According to Shin and Bianco [2020], users expect fairness, trust, transparency, and accountability, and these approaches can be used to meet these expectations. Transparency does not have to be jargon or heavily technical. Understanding what users' trust requirements are or what they are most concerned about can lead to what design solutions are needed [8,9]. Proposed solutions include:

- **FAQs:** (Frequently Asked Questions): user-generated questions covering the most common issues as well as providing information on blockchain processes [10]
- **Videos:** short overviews (approximately 90 seconds) with comedic tones are encouraged [10]
- **Reputation ratings:** platform-based feedback dependent on the community of users [9]

FAQs and videos are familiar information tools users have likely used on other platforms [10]. Similarly, ratings and measuring trust based on the previous experience of other users are standard across multiple platforms and services. However, these design solutions may not be enough to build long-term trust or maintain continued use unless it is

incorporated into a company's broader goals and vision. Additionally, these proposed solutions have not been validated to address error prevention.

EXISTING STRATEGY RECOMMENDATIONS

According to Saez (2020), developing a strategy is the first step for a company team after creating an initial concept. Plansky, O'Donnell, & Richards ' (2016) strategy guide for blockchain-enabled platforms includes the following:

- **Scope:** defining and focusing on specific opportunities or potential starting points
- **Assessing feasibility and readiness:** form a hypothesis for how blockchain can be effective in the business which should narrow the list of potential starting points
- **Prototyping & testing:** adjust parameters for prototypes, maintain the hypothesis throughout iterations, and include team members who understand the company's business strategy
- **Scaling:** create long-term plans based on the results of the initial prototypes, such as increasing revenue and quality improvement

Rayemaker's [2015] recommendations align with traditional business strategy recommendations but do not consider user trust, experience, or design. Alternatively, based on interviews and observations with blockchain leaders and stakeholders of blockchain and platform strategy, Saez [2020] provides a comprehensive model for every phase of development (see Figure 2). It incorporates Raymaekers' recommendations but also discusses security. This is important because security is often not addressed in design when considering cryptocurrency [1,7]. However, Saez [2020] recommends detailed attention to operational security practices. There is minimal consideration of usability and addressing user challenges since these recommendations are based on data from leadership stakeholders.

Although agile practices and design iterations are included, they only appear in the final phase of piloting. Additionally, agile prioritizes problem-solving with a quick turnaround for solutions. Still, trust-building through UX is yet to be considered a high priority [4]. There is a struggle with a lack

of consensus on its value and it may not fit into an agile sprint[4].

EXPANDING STRATEGY CONSIDERATIONS

The path forward would be to merge existing approaches to strategy and UX design solutions to enhance trust. Rather than solely focusing on functional attributes of UX, hedonic attributes or subjective experiences such as user trust need to be incorporated into multiple phases of product development and business strategy to provide shared goals and language across teams [4]. Similarly, UX professionals should be involved in projects early with continued roles throughout each phase of development or design [4]. There are competing goals between UX and non-UX practitioners, such as developing efficient code vs. better designs [4]. However, building a user-centered strategy creates an awareness of UX goals and their implications which can incorporate be into guidelines, tools, and methods as daily practices [4].

CONCLUSION & FUTURE WORK

Due to the volatile nature of cryptocurrency markets, a lack of regulation, and a lack of knowledge from managers of blockchain platforms, trust is a major factor in adopting cryptocurrency. Building trust through design provides users, especially novices, an accessible way to engage with cryptocurrency comfortably. As an alternative to waiting for and lobbying for regulation, creating a strategy that emphasizes the importance of trust to UX maintains the decentralization goal of cryptocurrency. It also meets practical business requirements of meeting user needs and expectations.

This paper has shown the significance of trust in user experience and strategy. Understanding how companies can improve trust for cryptocurrency adoption has implications for other blockchain applications such as voting systems [5] and smart contracts[9]. Future research will focus on creating a UX strategy model specific to cryptocurrency from a user-centered perspective. Following the creation of the model, a case study that incorporates stakeholder and user feedback will be the next step. Additional factors that can shape future work include how stable established financial institutions and payment services such as banks and PayPal have integrated blockchain and if these integrations have increased user trust among standard users or users who value decentralization.

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